

Chapter 1: Introduction to IT Infrastructure

Objective

- Explain IT Infrastructure Architecture
- Discuss IT building blocks
- Discuss Non-functional attributes
- Discuss Cloud Computing
 - Discuss the cloud characteristics
 - Explain the cloud deployment models
 - Explain the cloud service models
 - Explain Infrastructure as a Service (IaaS)
 - Explain edge computing

Introduction to IT Infrastructure Architecture

Introduction

- IT infrastructures become more complicated due to new types of applications:



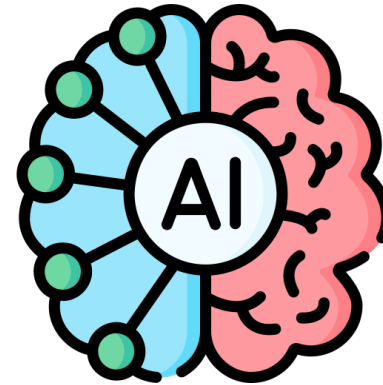
Mobile
Computing



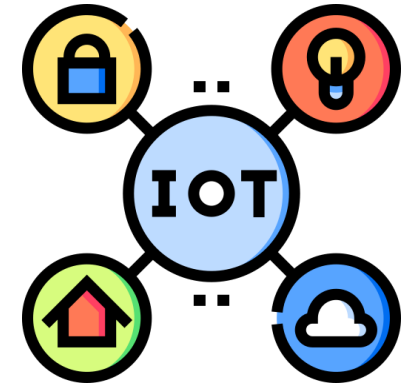
Cloud
Computing



Big Data



Artificial
Intelligence



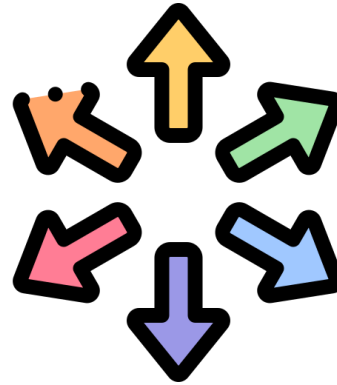
Internet of
Things

Introduction

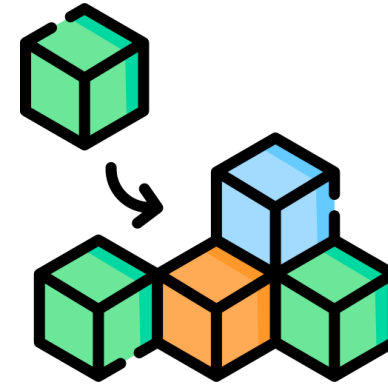
- Agile adaptations require infrastructure:



Solid



Scalable



Modular

The definition of IT infrastructure

- In literature, many definitions of IT infrastructure are described:



WIKIPEDIA

"(IT) components that are the foundation of an IT service; typically physical **components** (computer and networking hardware and facilities), but also various **software** and **network** components. "

The definition of IT infrastructure



“All of the **hardware, software, networks, facilities**, etc., that are required to develop, test, deliver, monitor, control, or support IT services.

The term IT Infrastructure includes all of the Information Technology but not the associated people, processes and documentation.”

The definition of IT infrastructure



“IT infrastructure refers to the composite **hardware, software, network** resources and **services** required for the existence, operation and management of an enterprise IT environment.

IT infrastructure allows an organization to deliver IT solutions and services to its employees, partners and/or customers and is usually internal to an organization and deployed within owned facilities. “

The definition of IT infrastructure



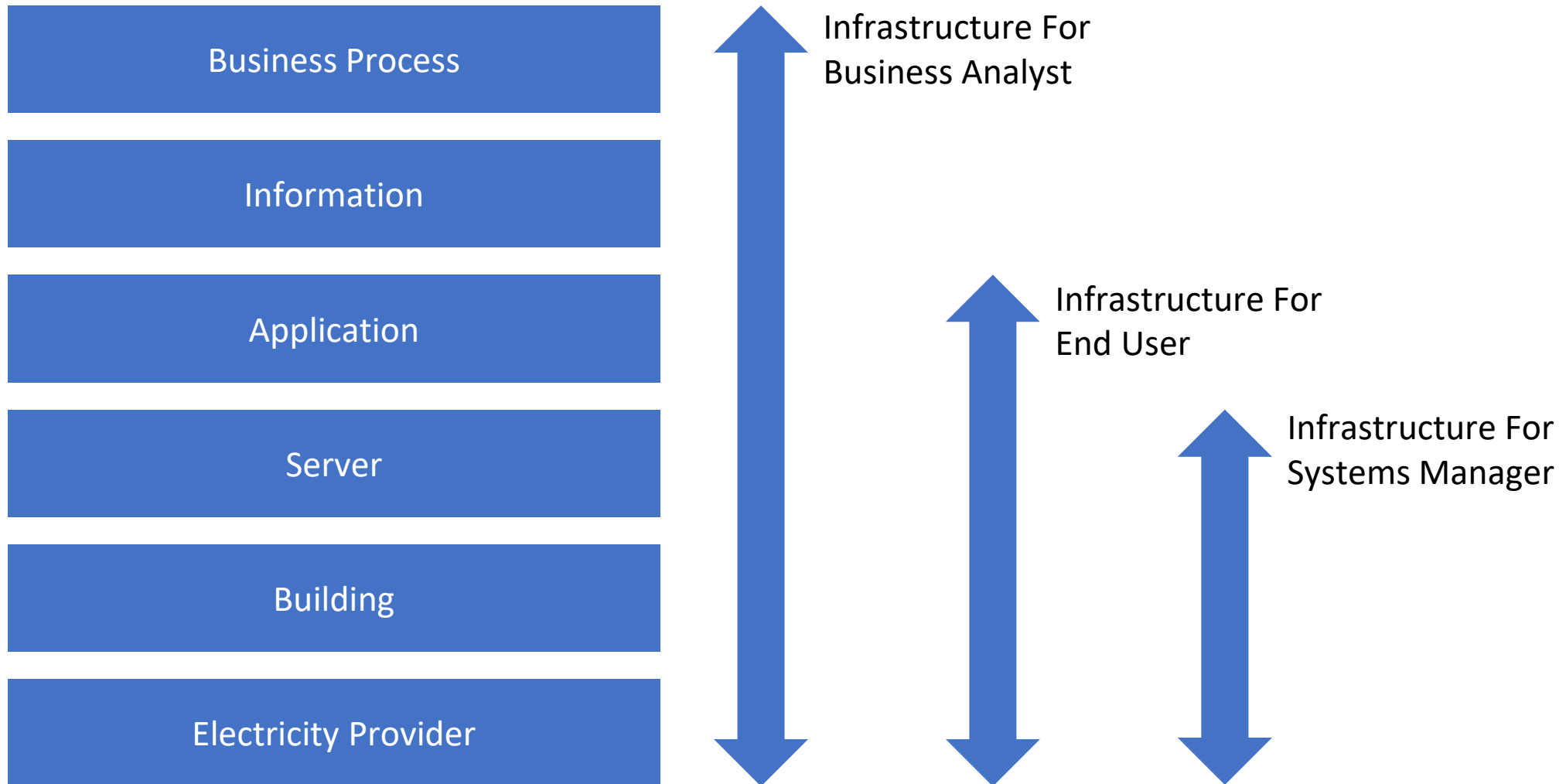
“IT infrastructure is the system of **hardware, software, facilities** and **service** components that support the delivery of business systems and IT-enabled processes.”

The definition of IT infrastructure



“IT infrastructure refers to the combined components needed for the operation and management of enterprise IT services and IT environments.”

The definition of IT infrastructure



The definition of IT infrastructure

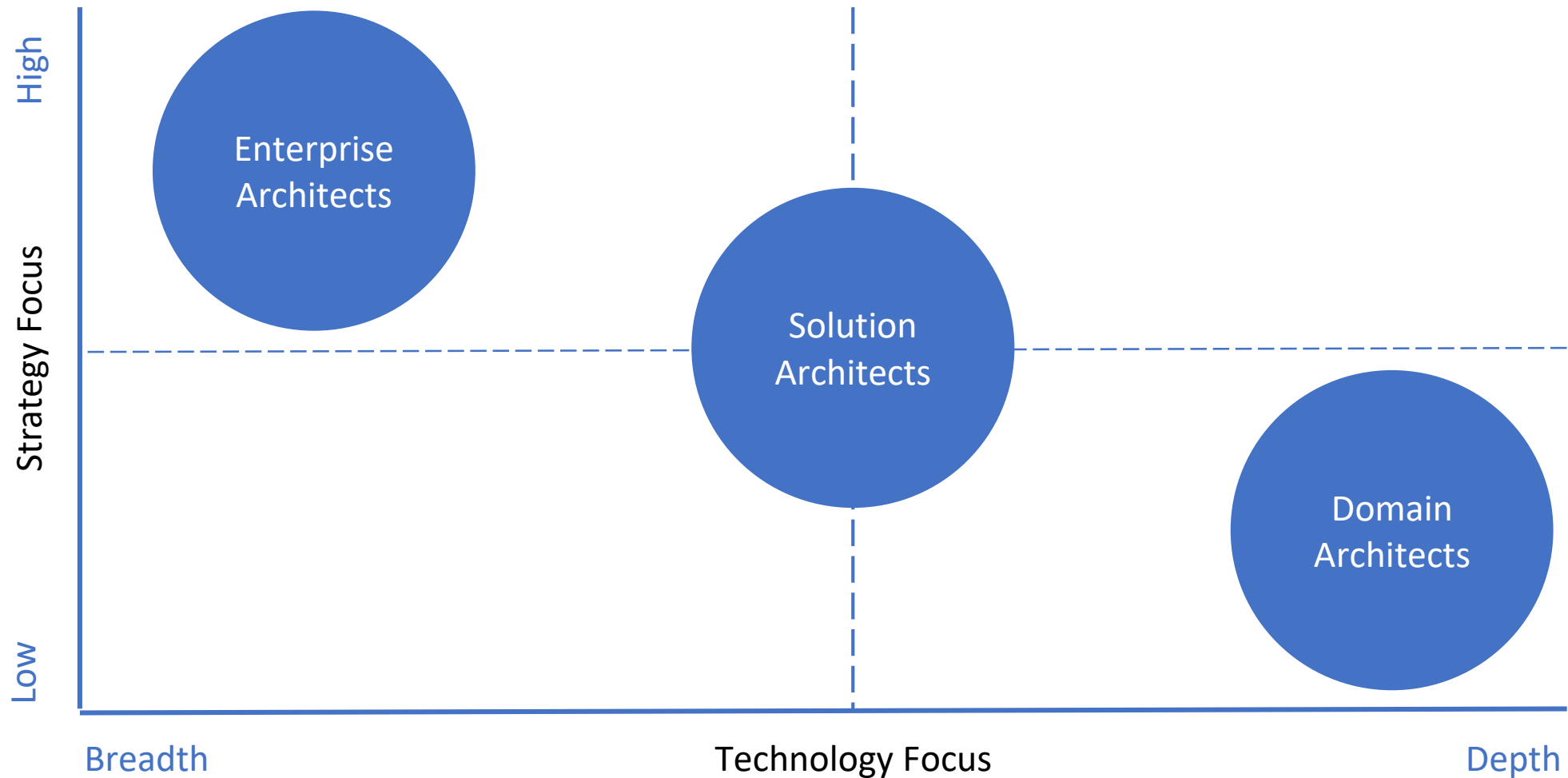
- What infrastructure comprises depends on:
 - Who you ask
 - What their point of view is
- For most people, infrastructure is invisible and taken for granted

Introduction

- Architecture is the philosophy that underlies a system and defines its **purpose, intent, and structure**.
- Architecture is crucial to control the infrastructure when it is:
 - Designed
 - In use
 - Changed



What is IT architecture?



“To empower every person and every organization on the planet to achieve more.”

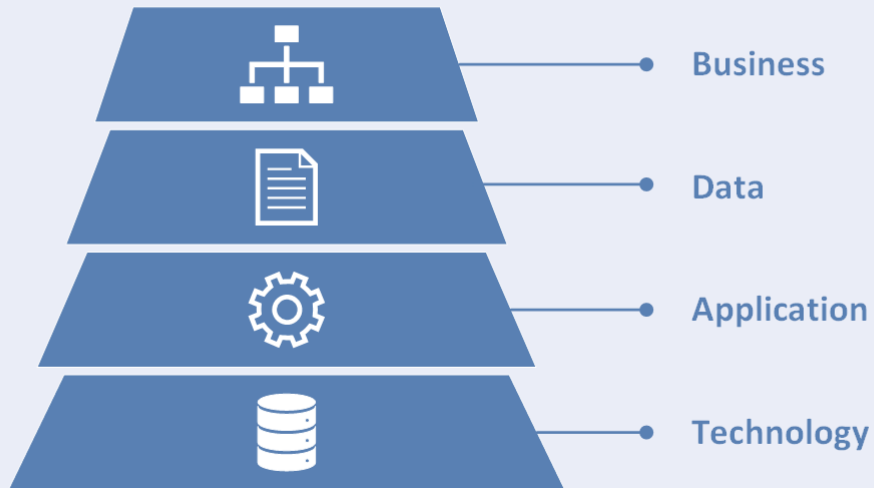


“Utilize the power of Moore's Law to bring smart, connected devices to every person on earth.”



What is IT architecture?

Domains of Enterprise Architecture

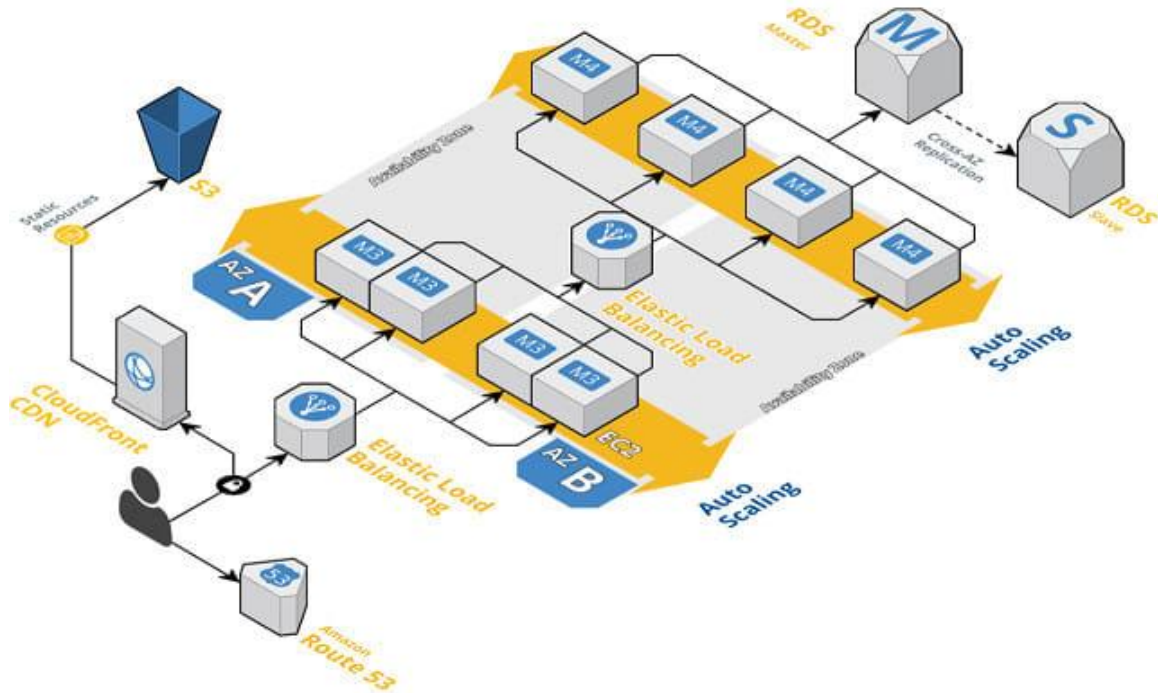


Domain architects

- They are **experts** on a particular **business or technology topic**.
- They most often work for **infrastructure or software vendors**.
- They assist solution architects on projects.

What is IT architecture?

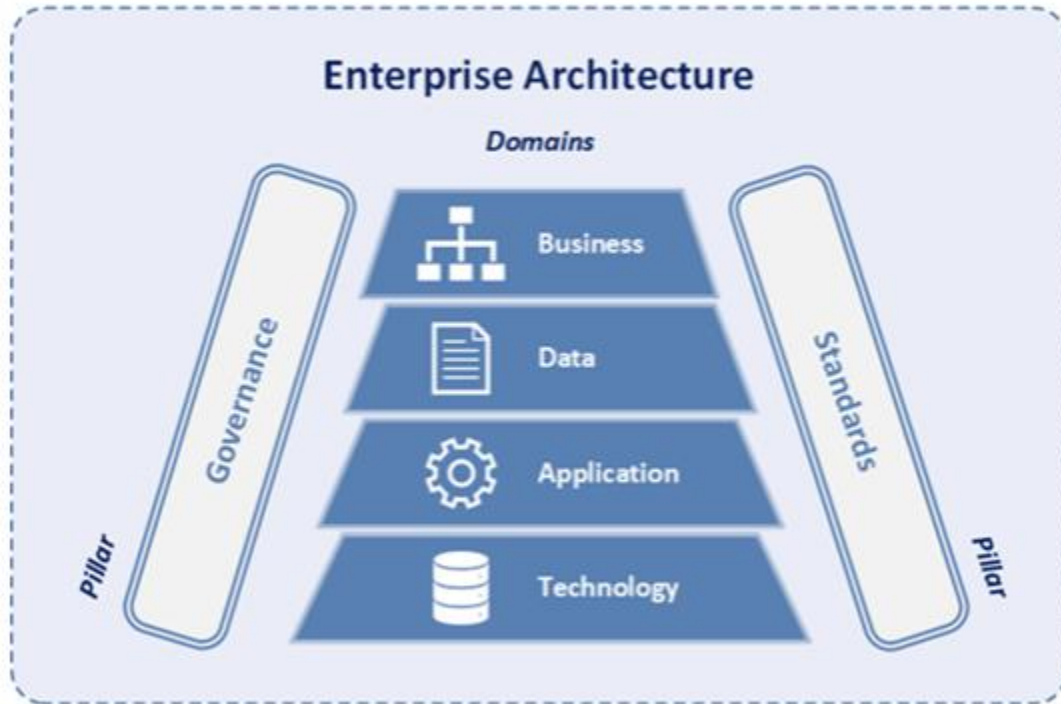
AWS Certified Solution Architect



Solution architects

- They create **IT solutions**.
- They are the **technical conscience** and authority of a project.
- They responsible for **architectural decisions** in the project.
- They work closely with the project manager.

What is IT architecture?

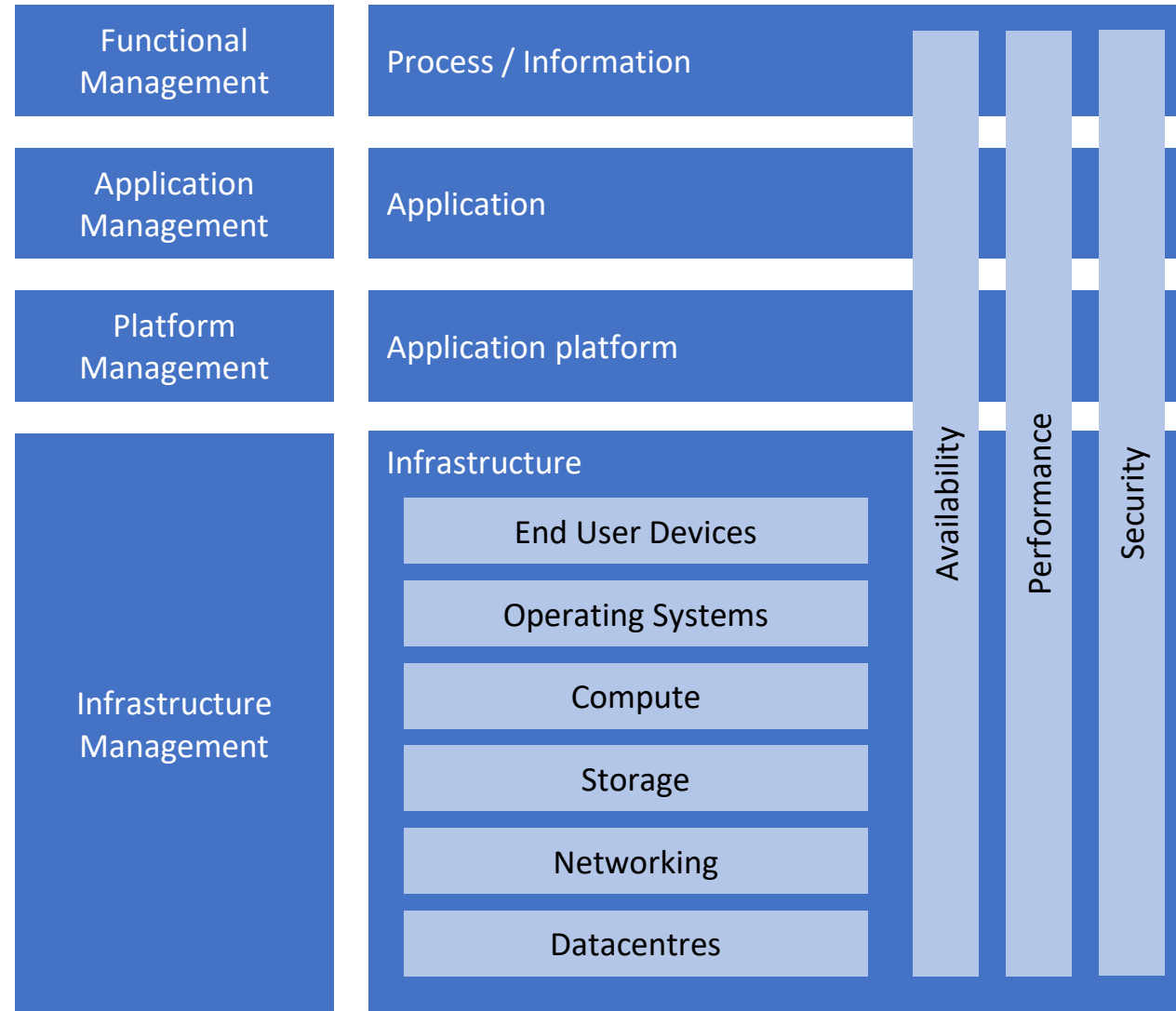


Enterprise architects

- They continuously **align an organization's entire IT landscape** with the **business activities** of the organization.
- They typically work closely with the CIO and business units to align the needs of the business with the current and future IT landscape.

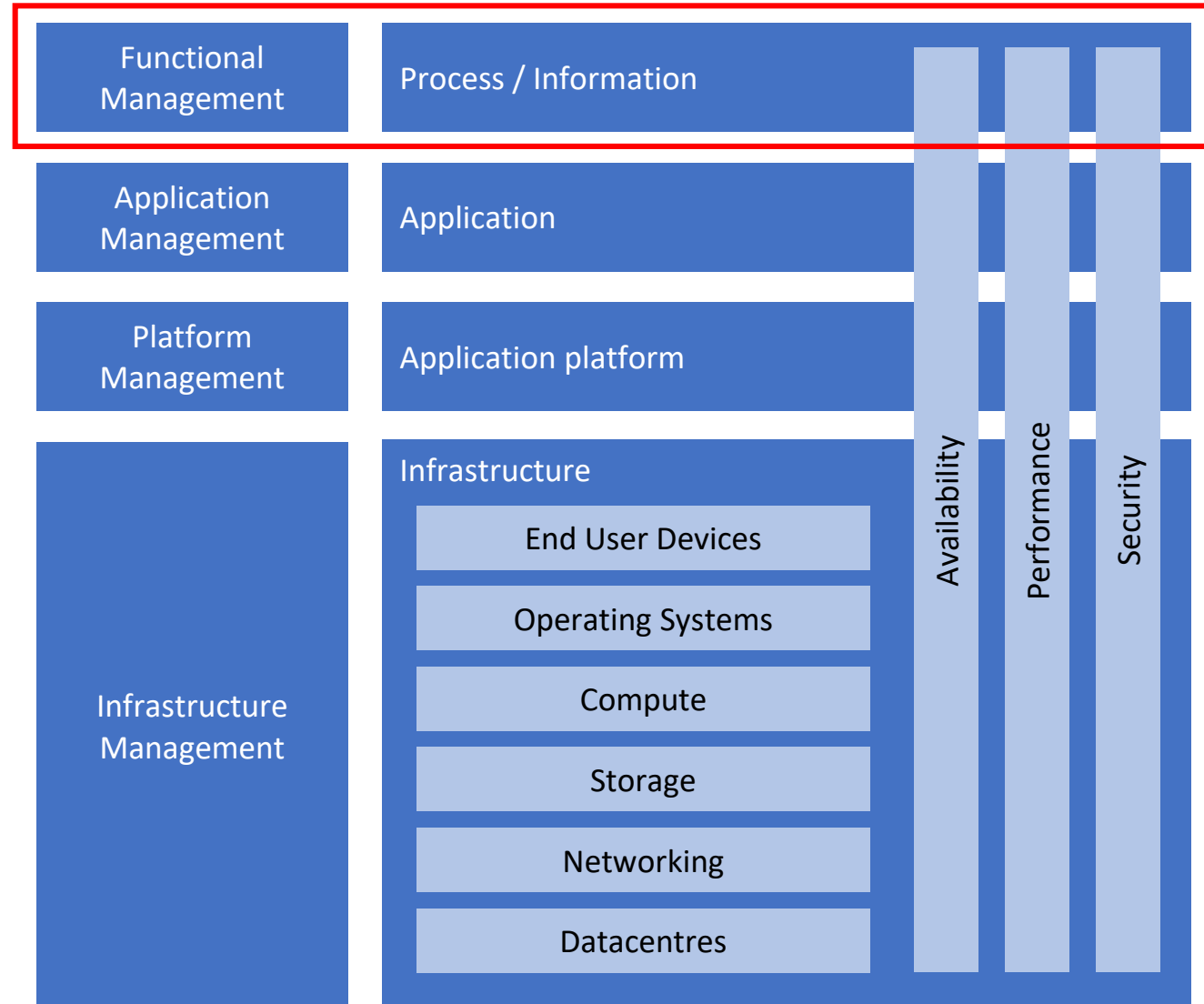
IT building blocks

IT building blocks



Processes / Information building block

- Organizations implement business processes to fulfil their mission and vision
- Processes are organization specific
- Business processes create and use information



Management

```
graph TD; Management[Management] --- Operations[Operations<br/>Production<br/>Supply Chain]; Management --- Finance[Finance<br/>Accounting<br/>Procurement<br/>HR]; Management --- Marketing[Marketing/Sales<br/>Marketing<br/>Sales<br/>Customer Service]; Management --- RnD[R&D<br/>Development<br/>IT];
```

Operations

Production
Supply Chain

Finance

Accounting
Procurement
HR

Marketing/Sales

Marketing
Sales
Customer Service

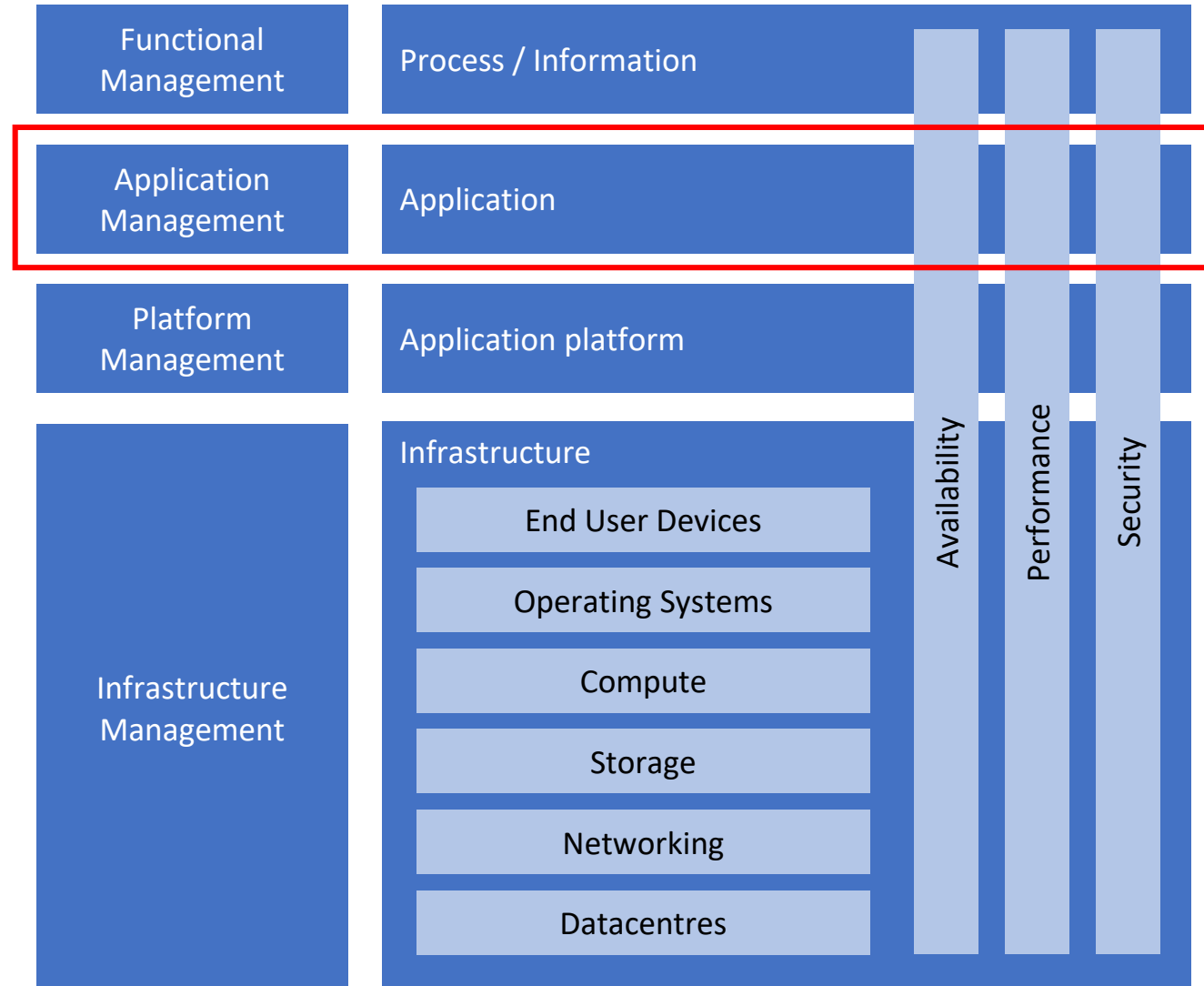
R&D

Development
IT

Applications building block

The Applications building block includes several types of applications based on the following characteristics:

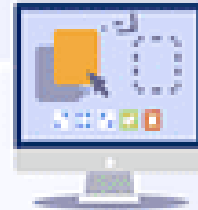
- **Usage:** Applications can be single-user or multi-user.
- **Source:** Applications can be purchased as commercial off-the-shelf (COTS) products or developed as custom software.
- **Architecture:** Applications can be designed as standalone applications or as multi-tier applications.
- **Timeliness:** Real-time systems, are used in manufacturing, logistics, or other environments where timeliness is critical. Interactive applications respond to user actions, such as mouse clicks. Batch-based systems that process data for hours at a time.



Cases of Enterprise Business Applications



**Enterprise Resource
Planning (ERP)**



**Business
Intelligence
Systems**



**Customer
Relationship
Management
(CRM)**



**Point-of-Sale
(POS) software**



**Human Resource
Systems (HR)**



**Accounting and
billing systems**

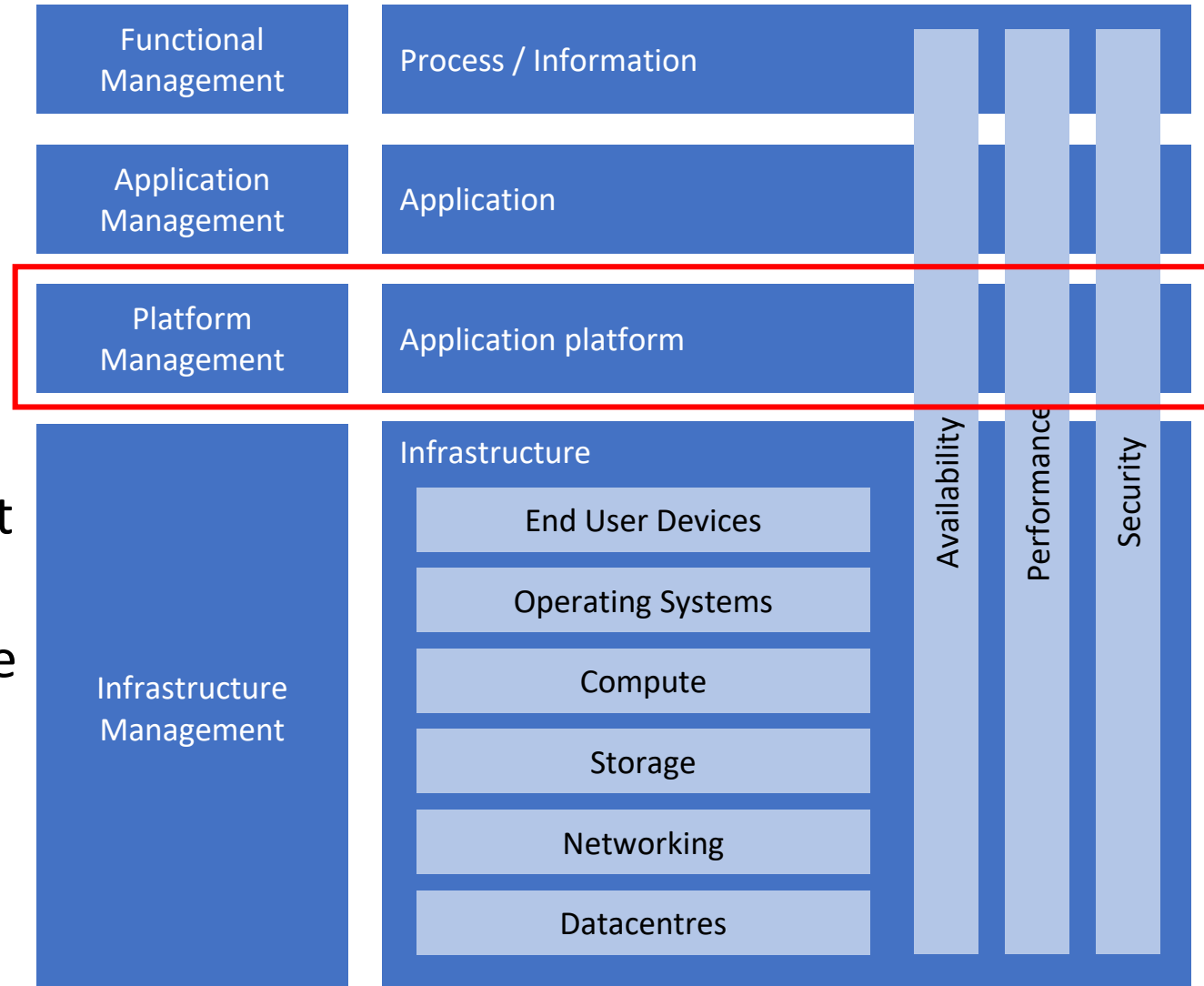


**Supply chain
management
(SCM)**

Application Platform building block

We can identify the following services as part of the application platform building block:

- **Application servers** provide services to applications.
- **Container platforms** like Kubernetes, Azure Container Instances, and Amazon Elastic Container Service, that run docker containers.
- **Connectivity** entails Enterprise Service Buses (ESBs) like Microsoft BizTalk, the TIBCO Service Bus, IBM MQ, and SAP NetWeaver PI.
- **Databases** provide a way to store and retrieve structured data.

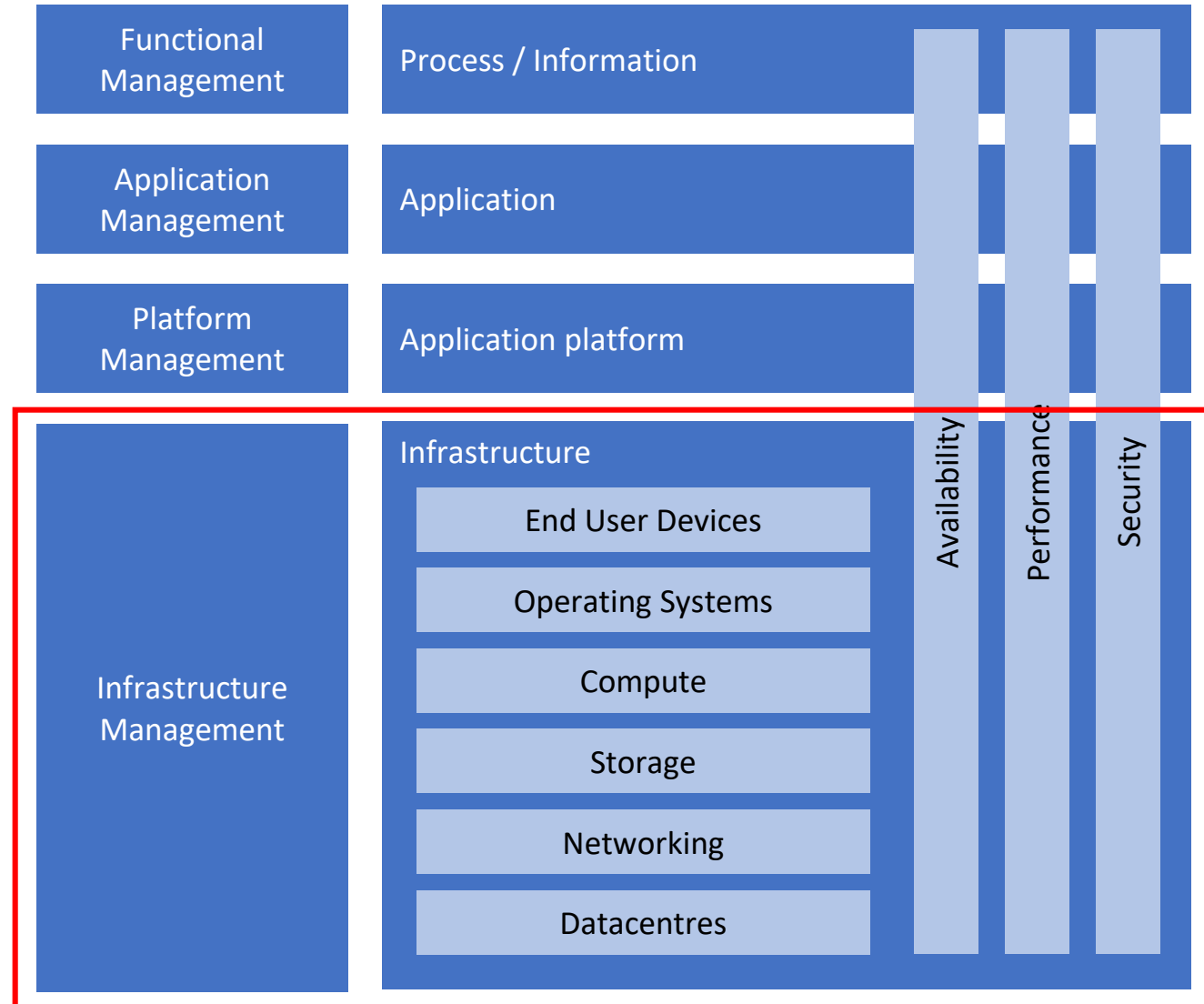


Application Platform building block

- Front-end servers provide end users with interactions to applications
 - presenting application screens in web browsers
- Application servers act as containers running the actual application
- Connectivity entails File transfer protocol (FTP) servers, Extract, transform, and load (ETL) servers, and Enterprise Service Bus (ESB)s
- Databases provide a way to store and retrieve structured data

Infrastructure building blocks

- Comprises:
 - End User Devices
 - Operating Systems
 - Compute
 - Storage
 - Networking
 - Datacenters



Infrastructure building blocks

- End User Devices are the devices used by end users to work with applications
 - PCs
 - Laptops
 - Thin clients
 - Mobile devices
 - Printers



Infrastructure building blocks

- Operating Systems are collections of programs that manage a computer's internal workings:
 - Memory
 - Processors
 - Devices
 - File system



Infrastructure building blocks

- Compute are the physical and virtual computers in the datacenter. Also known as servers



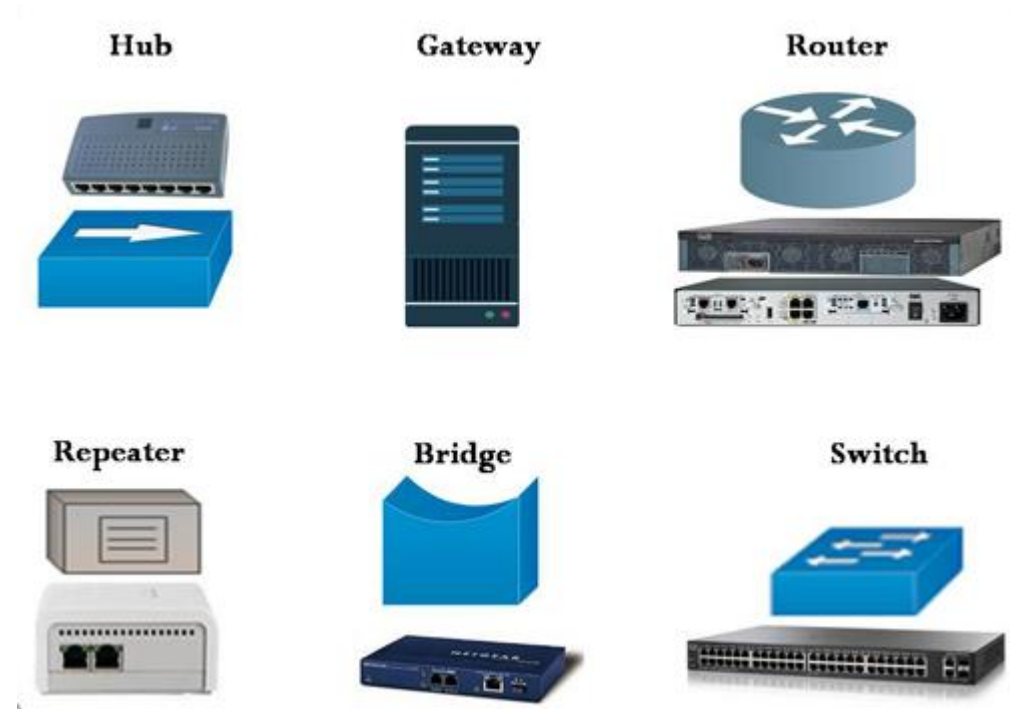
Infrastructure building blocks

- Storage are systems that store data
 - Hard disks
 - Tapes
 - Direct Attached Storage (DAS)
 - Network Attached Storage (NAS)
 - Storage Area Networks (SANs)



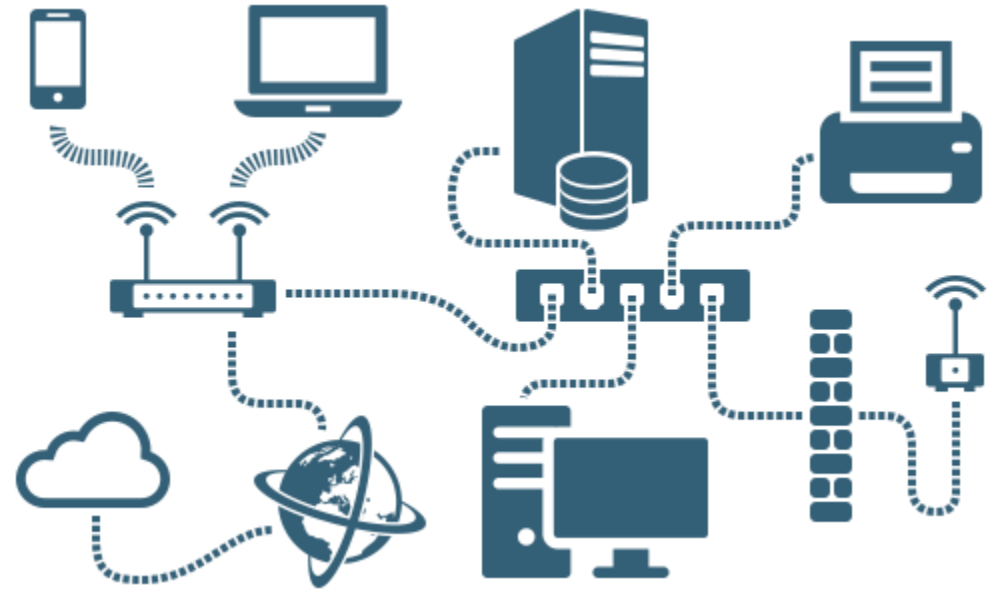
Infrastructure building blocks

- Networking connects all components
 - Routers
 - Switches
 - Firewalls
 - WAN
 - LAN
 - Internet access
 - VPNs



Infrastructure building blocks

- Includes infrastructure services
 - DNS
 - DHCP
 - Time services



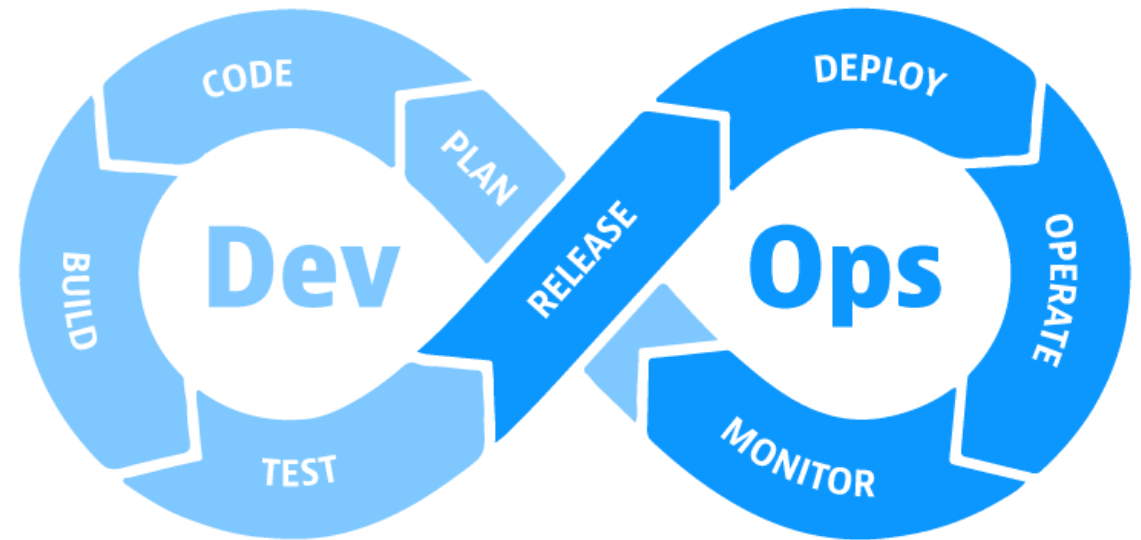
Infrastructure building blocks

- Datacenters are locations that host most IT infrastructure hardware
 - Uninterruptible power supplies (UPSs)
 - Heating, Ventilation, and Air Conditioning (HVAC)
 - Computer racks
 - Physical security measures



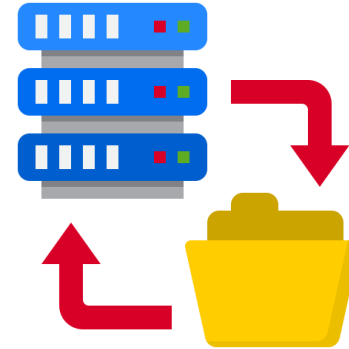
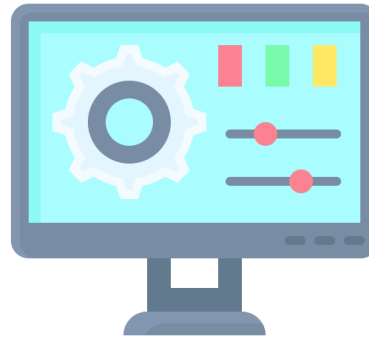
Infrastructure building blocks

- Infrastructure management are processes
 - Information Technology Infrastructure Library (ITIL)
 - Control Objectives for Information and Related Technology (COBIT)
 - DevOps



Infrastructure building blocks

- Tools are used for:
 - Monitoring
 - Backup
 - Logging



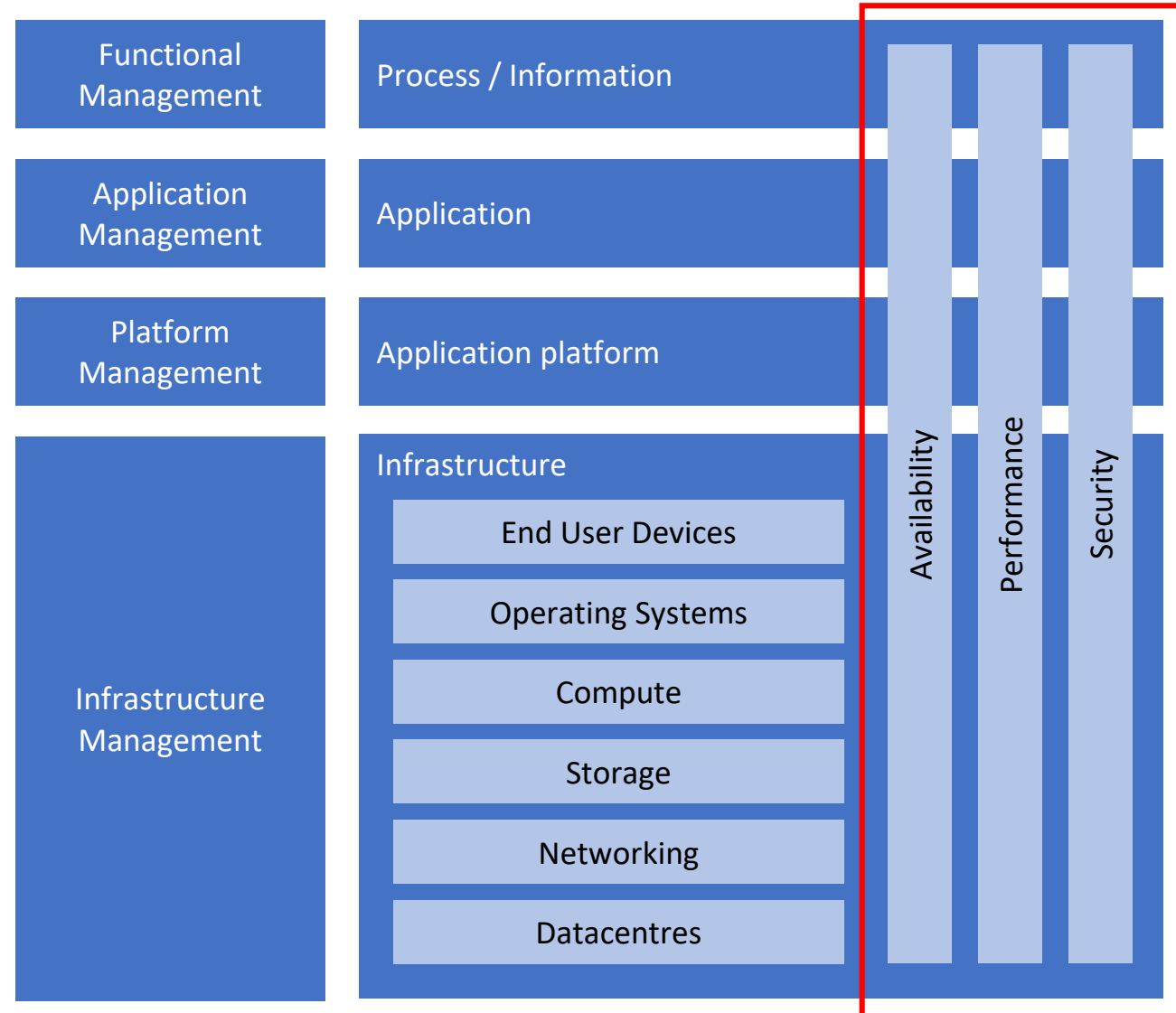
Infrastructure building blocks

- Infrastructure building blocks are not per definition hierarchically related!
 - For instance, servers need both networking and storage
 - Both are equally important

Non-functional attributes

Non-Functional attributes

- Non-functional attributes describe the qualitative behavior of a system
- Essential in IT infrastructure architectures:
 - Availability
 - Performance
 - Security



NFRs

- The name “Non-functional attributes” suggests they have no function
- They are very important for the successful implementation and use of an IT infrastructure
- The term non-functional requirements or NFRs is frequently used and widely known
- The acceptance of a system is largely dependent on the implemented non-functional requirements

Conflicting NFRs

- Many of the non-functional attributes are delivered by the infrastructure
- Non-functional requirements are often conflicting:
 - Security versus user friendliness
 - Performance versus cost
- The infrastructure architect should present stakeholders with these conflicting requirements and their consequences

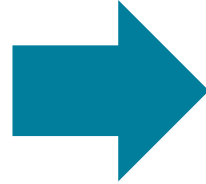
Cloud computing and infrastructures

Cloud Computing

Introduction

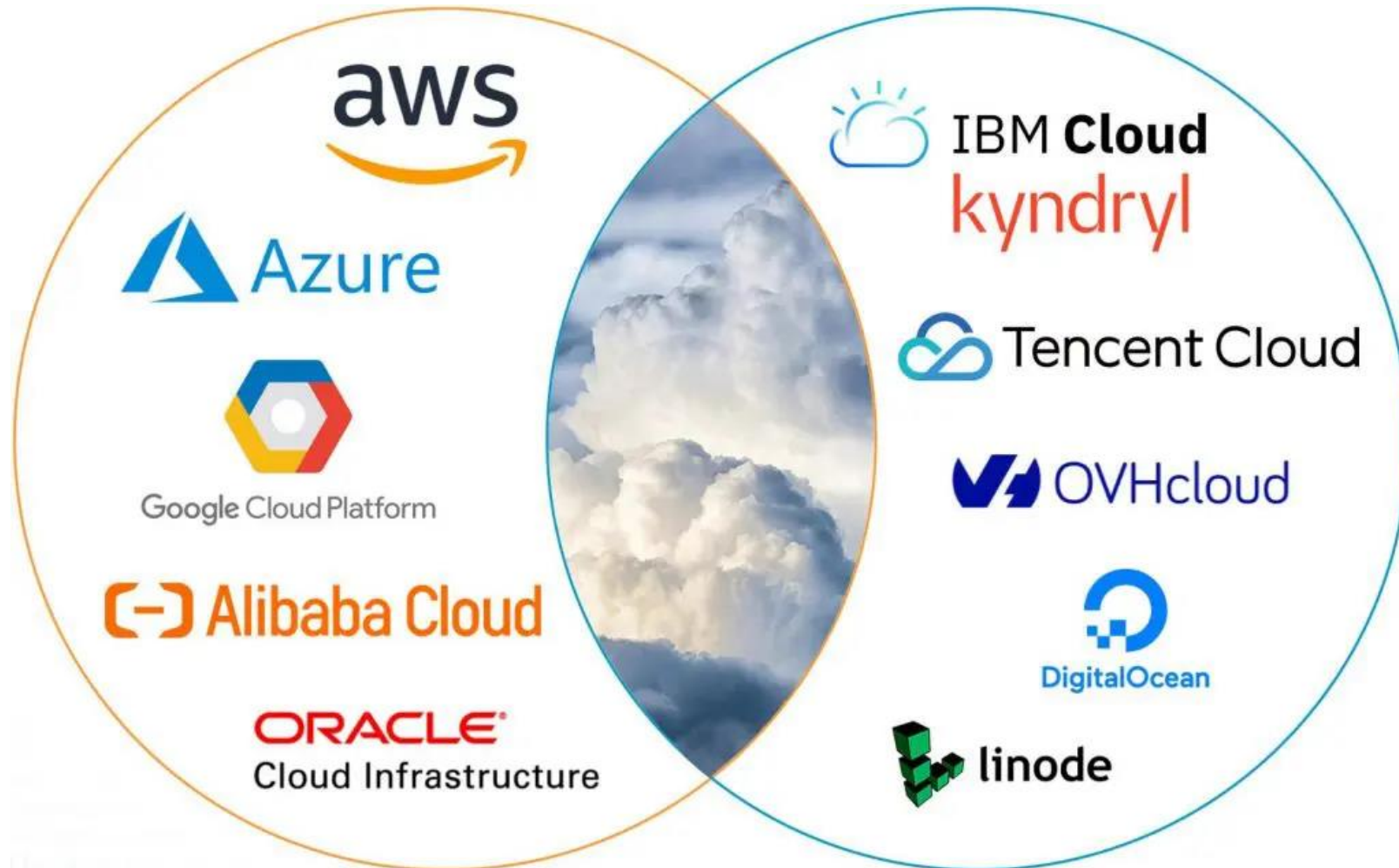


ON-PREMISE



CLOUD

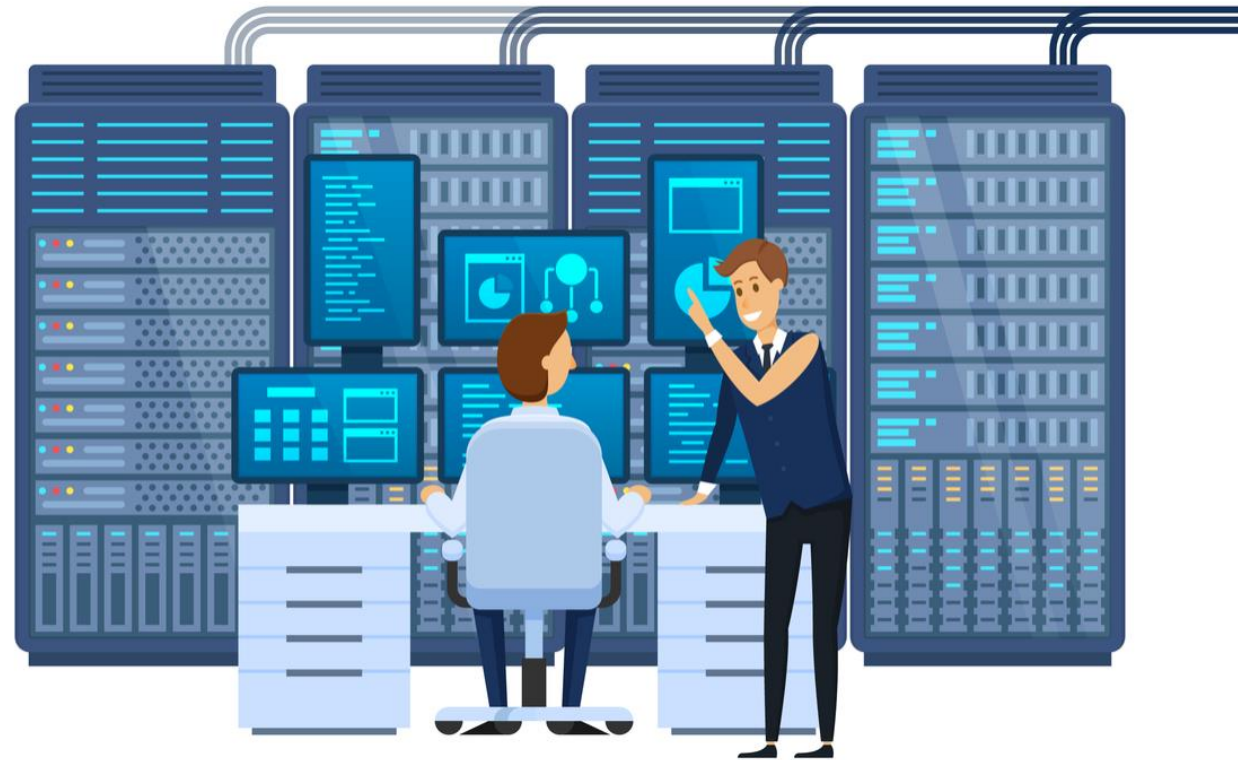
Public cloud providers



Source: <https://dgtlinfra.com/top-cloud-service-providers/> (18 April 2024)

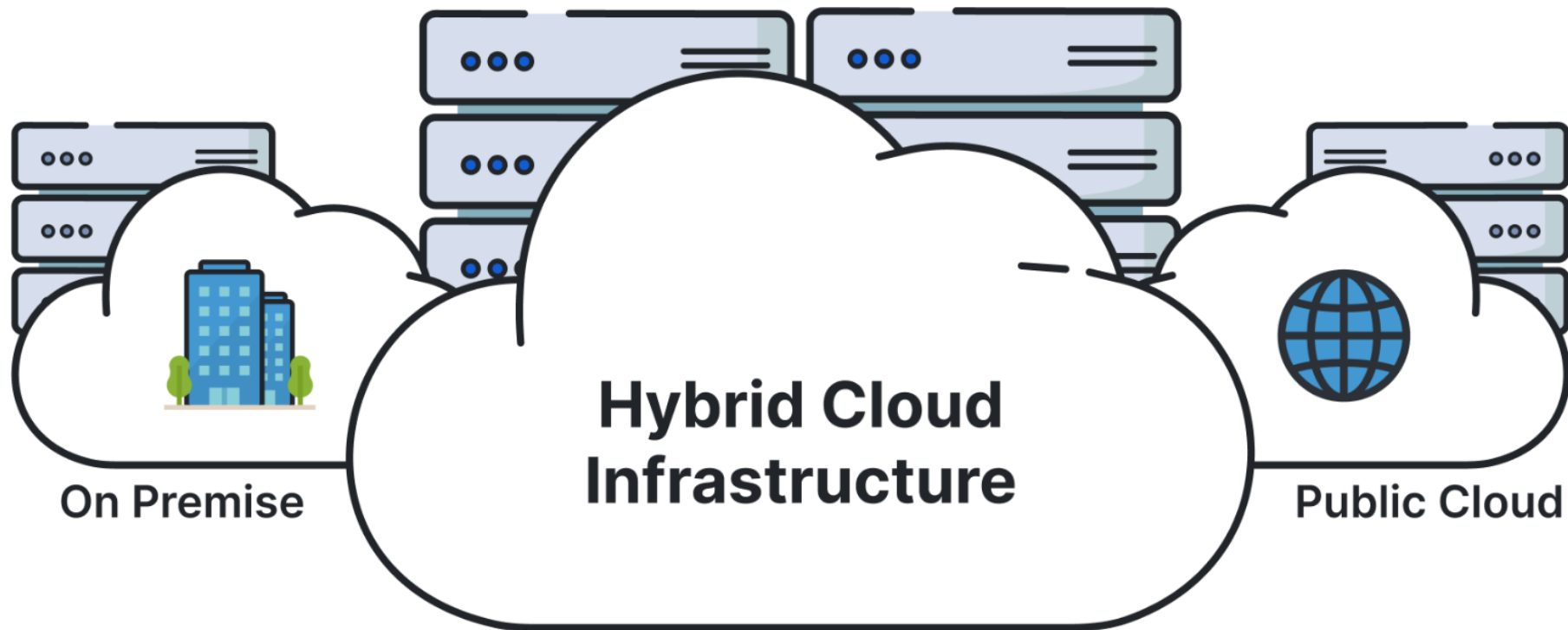
Cloud is the new infrastructure

Most organizations will be using on-premises infrastructure for many years to come



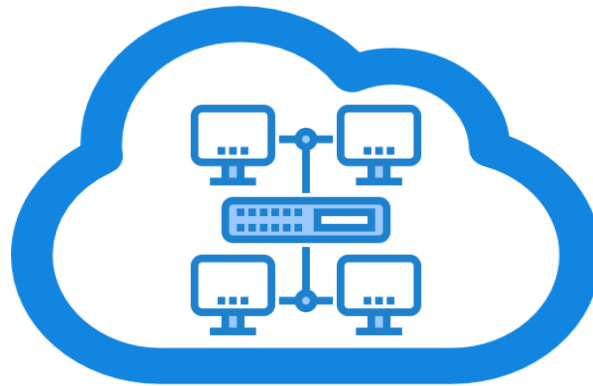
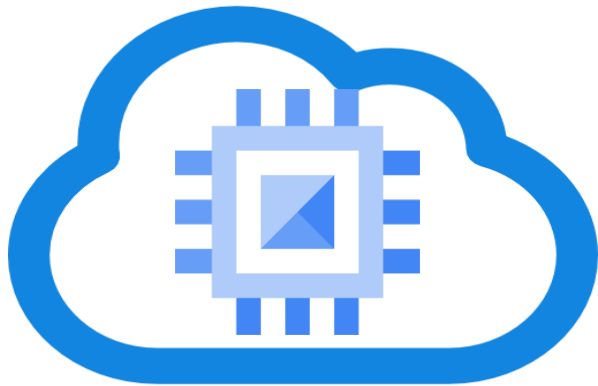
Cloud is the new infrastructure

In many cases, there will be a hybrid situation - on-premises + clouds



Cloud is the new infrastructure

The cloud is just a number of datacenters that are still filled with hardware – compute, networking and storage



Cloud definition

Cloud computing is a model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources(e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned** and released with **minimal management effort** or service provider interaction

Cloud definition

Cloud computing is an ***outsourcing business model***.

It enables organizations to **cut cost** while at the same time focusing on their primary business.



Cloud Characteristics

Cloud characteristics

On demand
self-service



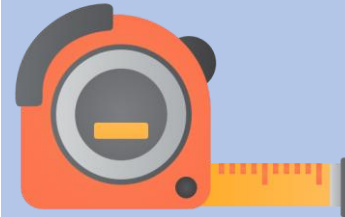
Rapid
Elasticity



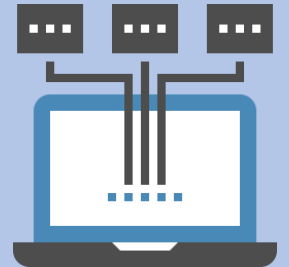
Resource
Pooling



Measured
service



Broad
network
access



Cloud characteristics

On demand
self-service



- Minimal systems management effort is needed to deploy systems or applications in a cloud environment.
- End users can configure, deploy, start and stop systems or applications on demand

Cloud characteristics

Rapid Elasticity



- A cloud is able to quickly scale-up and scale-down resources.
- When temporarily more processing power or storage is needed a cloud can scale-up very quickly on demand.
- When demand decreases, cloud resources can rapidly scale down, leading to elasticity of resources

Cloud characteristics

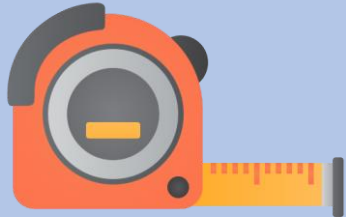
Resource Pooling



- Cloud computing provides applications with resources from a shared pool.
- This is typically implemented using virtualization technologies.

Cloud characteristics

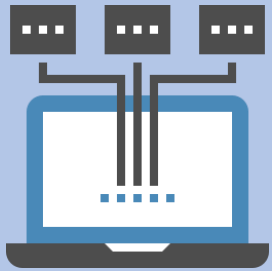
Measured
service



- The actual resource usage is measured and billed.
- There are no capital expenses, only operational expenses.
- This in contrast with the investments needed to build a traditional infrastructure.

Cloud characteristics

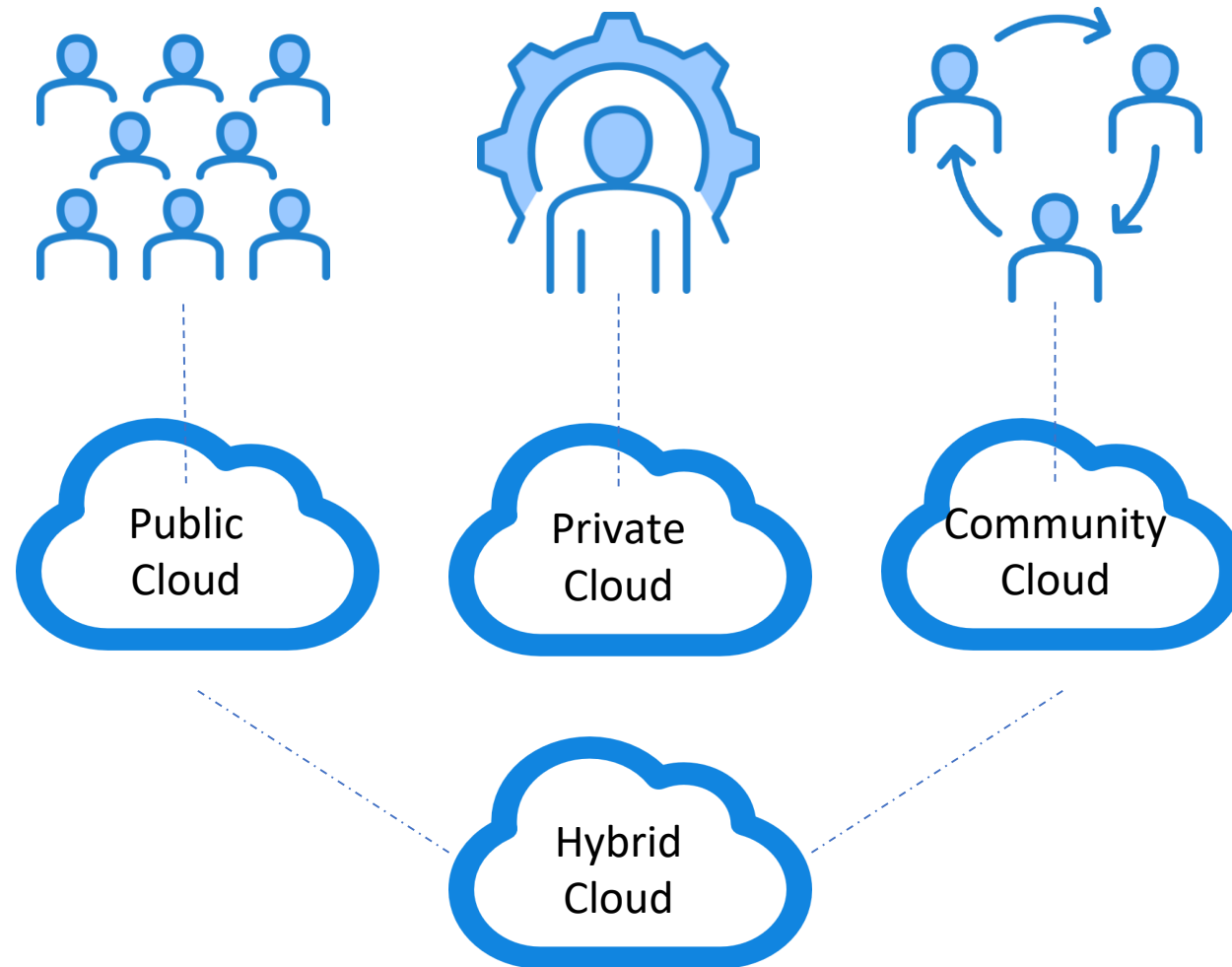
Broad
network
access



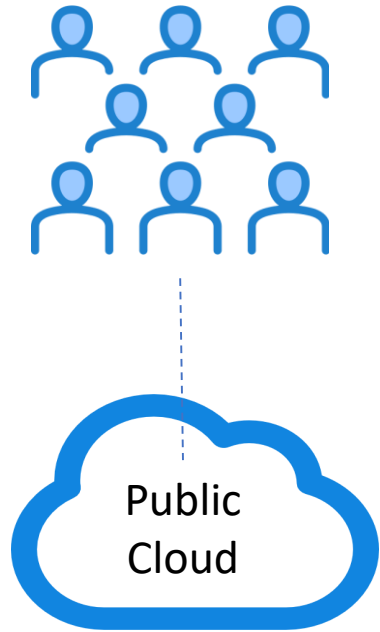
- Capabilities are available over the network and accessed through standard mechanisms.

Cloud Deployment Models

Cloud deployment models

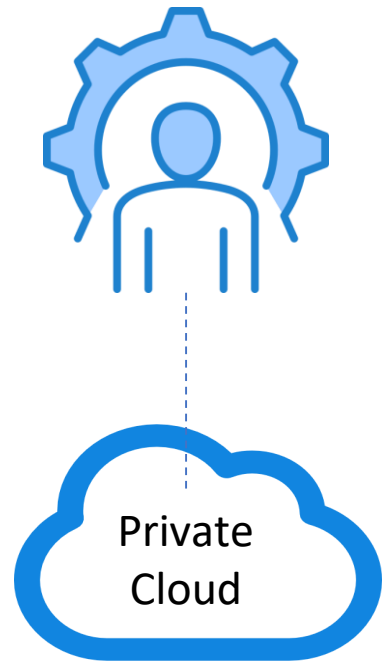


Cloud Deployment Models



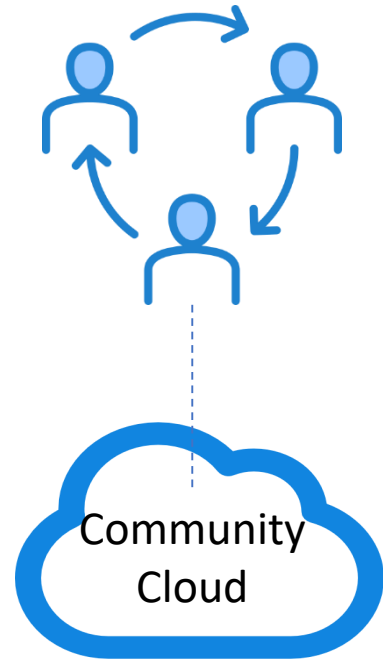
- **Public cloud** is delivered by a cloud service provider.
- It is accessible through the internet, and available to the general public.
- Public clouds largely benefit from economies of scale.

Cloud Deployment Models



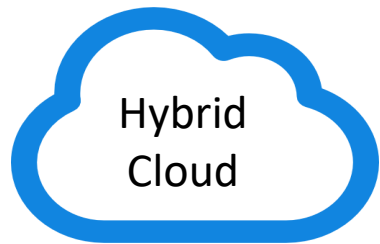
- **Private cloud** is operated solely for a single organization.
- It can be managed internally or by a third-party, and hosted either on premises or external.
- It extensively uses virtualization and standardization to bring down systems management cost and staff.

Cloud Deployment Models



- **Community cloud** is shared with a community of organizations that have shared concerns (like compliance considerations).
- It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination, and it may exist on or off premises

Cloud Deployment Models



- **Hybrid cloud** is provided by a combination of a public cloud, and a community cloud and/or a private cloud.
- This enables running generic services (like email servers) in the public cloud while hosting specialized services (like a business specific application) in the private or community cloud

Cloud Service Models

Cloud service models



SaaS

Software-as-a-Service

Consume



PaaS

Platform-as-a-Service

Build



IaaS

Infrastructure-as-a-Service

Host

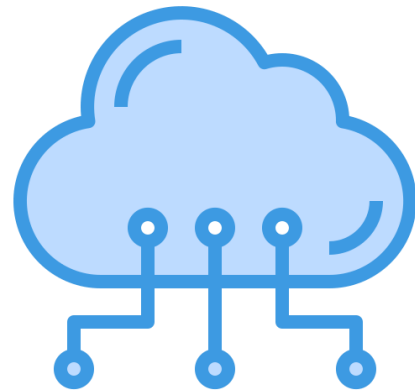
Cloud service models



Software-as-a-Service

- **Software-as-a-Service (SaaS)** delivers full applications that can be used by business users, and need little or no configuration.
- Examples are Microsoft Office365, LinkedIn, Facebook authentication, X authentication, and Salesforce customer relationship management (CRM)

Cloud service models

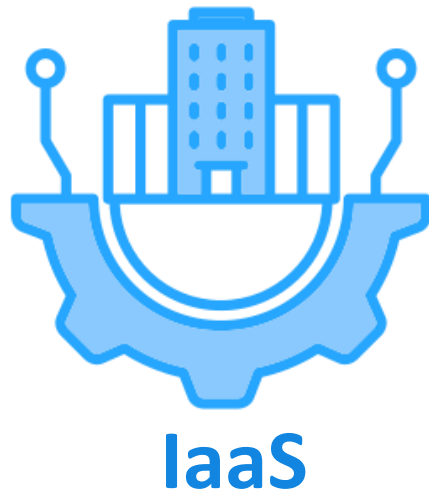


PaaS

Platform-as-a-Service

- **Platform-as-a-Service (PaaS)** delivers a scalable, high available, open programming platform that can be used by developers to build bespoke applications that run on the PaaS platform.
- Examples are Microsoft Azure Cloud Service and Google App Engine

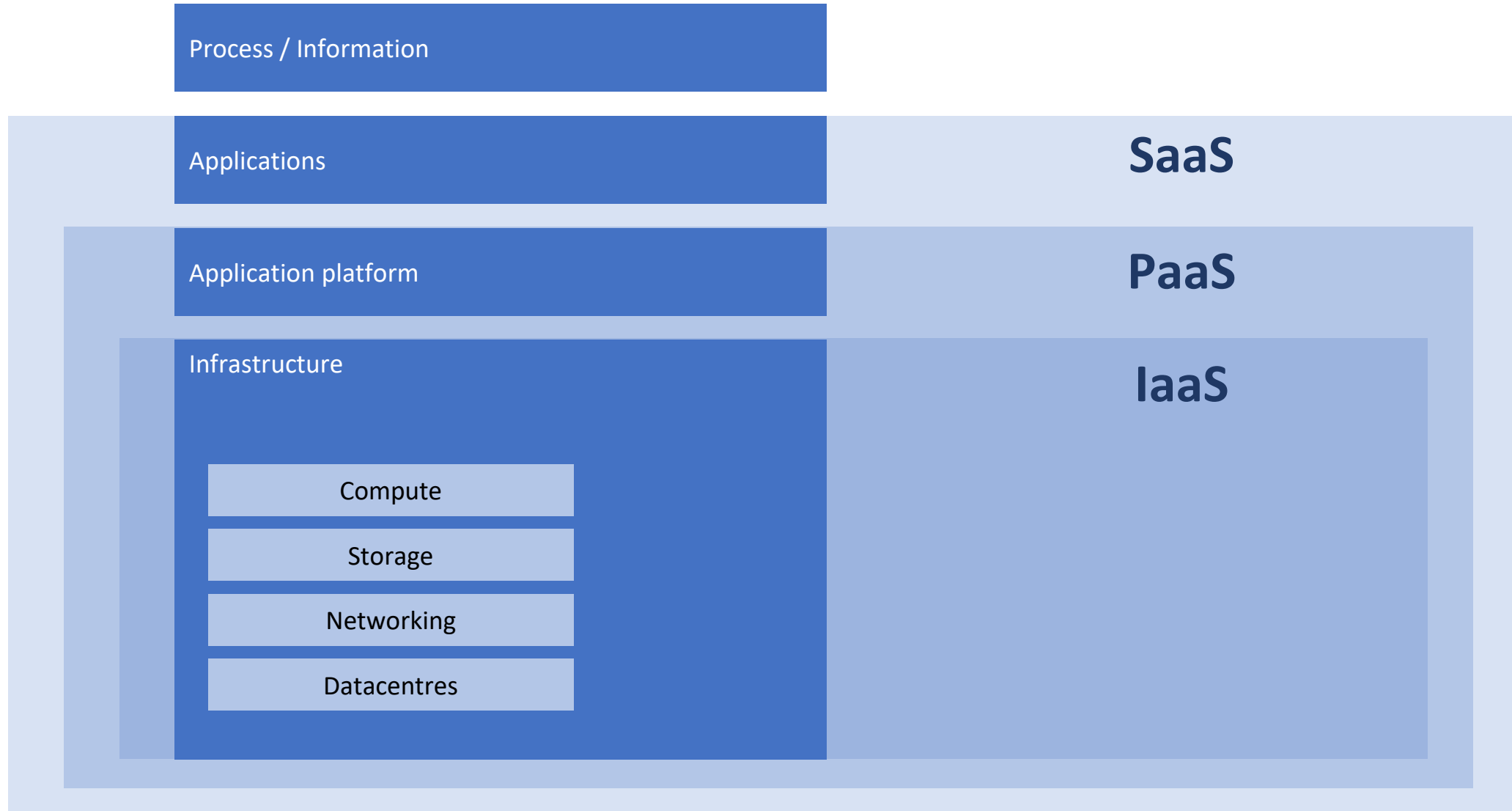
Cloud service models



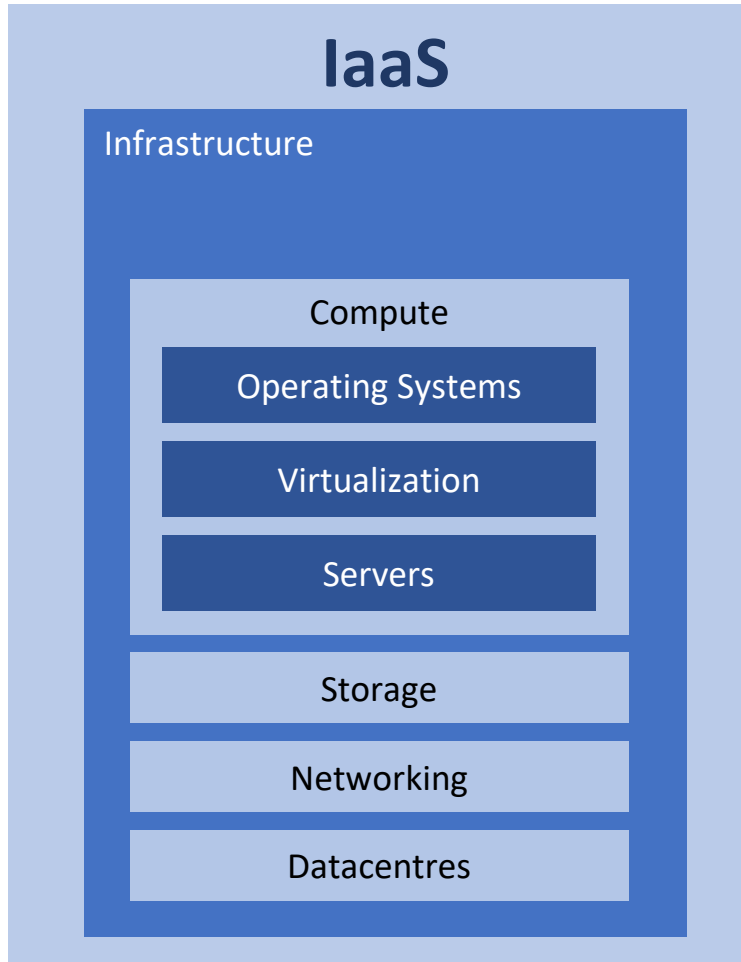
Infrastructure-as-a-Service

- **Infrastructure-as-a-Service (IaaS)** delivers virtual machines, networking, and storage. The user needs to install and maintain the operating systems and the layers above that.
- Examples are Amazon Elastic Cloud (EC2 and S3) and Microsoft Azure IaaS

Responsibilities

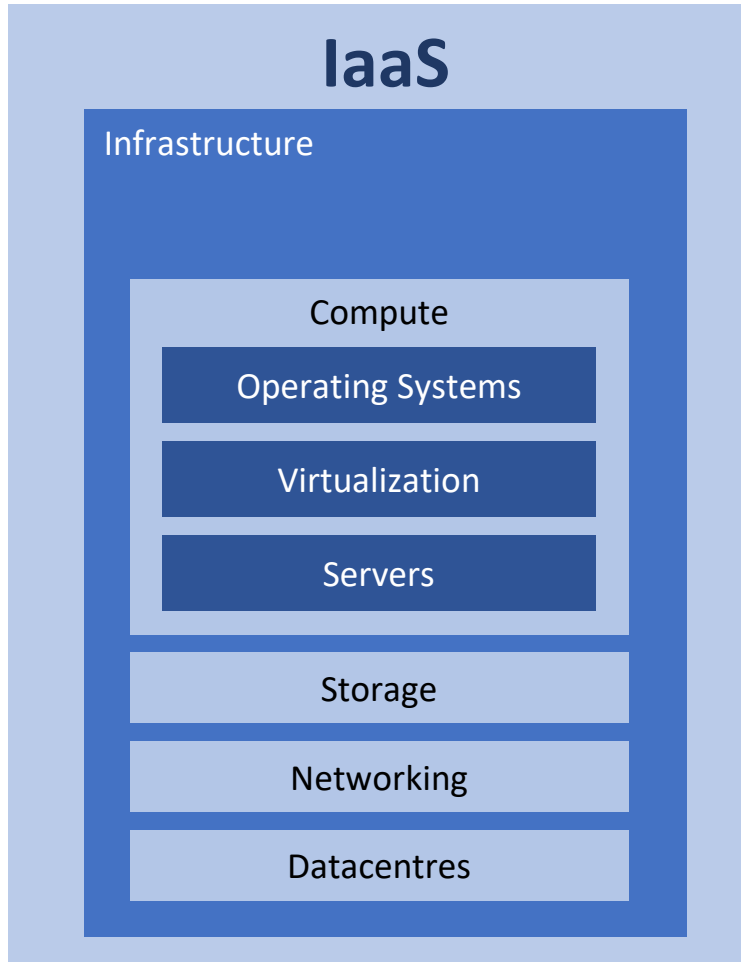


Infrastructure as a Service (IaaS)



- IaaS provides:
 - Virtual machines
 - Virtual storage
 - Virtual network
 - Management tools
- Applications running on IaaS should be robust to allow for failing hardware and should be horizontally scalable to increase performance

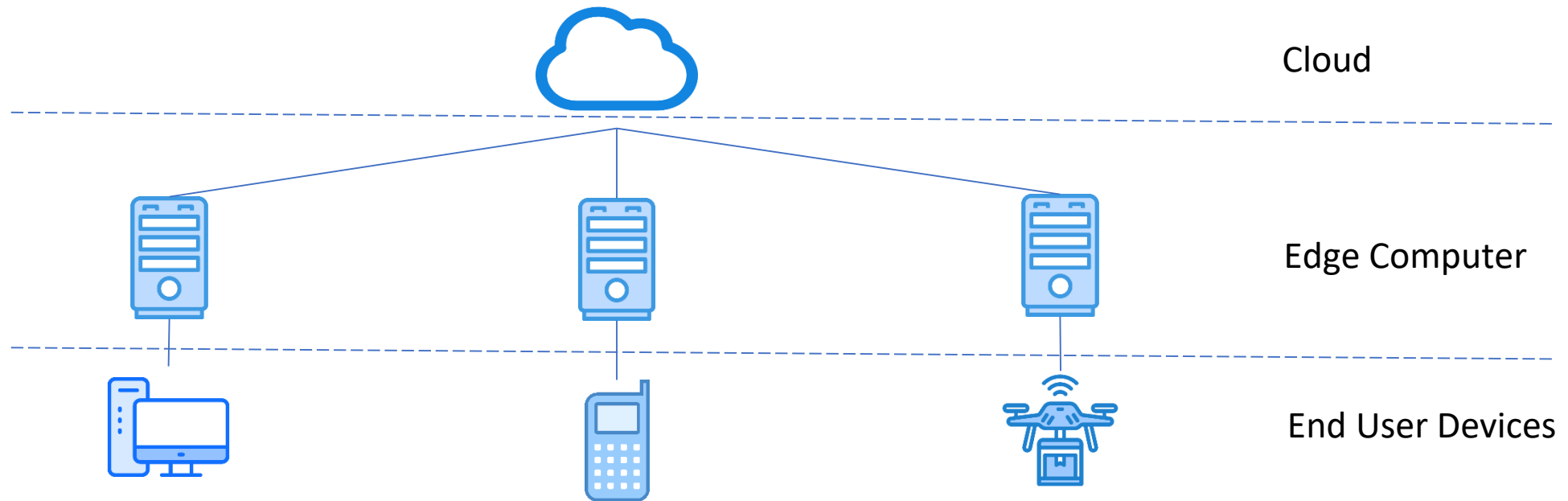
Infrastructure as a Service (IaaS)



- The user is responsible for patching and maintaining the operating systems and application software.
- Not all operating systems and applications can be used in an IaaS cloud; some software licenses prohibit the use of a fully scalable, virtual environment like IaaS.

Edge computing

- Edge computing brings computing power and data storage closer to where it is needed, rather than relying on a cloud or on-premises datacenter



Edge computing

- Computing and storage take place on devices **at the edge of the network**, such as routers, gateways, switches, and sensors
- Edge computing offers
 - Low latency
 - Low bandwidth needs
 - Real-time processing

Edge computing



- Edge computing is gaining popularity in Internet of Things (IoT) applications, where a large number of devices generate data that must be processed in real time

Conclusion

- Explain IT Infrastructure Architecture
- Discuss IT building blocks
- Discuss Non-functional attributes
- Discuss Cloud Computing
 - Discuss the cloud characteristics
 - Explain the cloud deployment models
 - Explain the cloud service models
 - Explain Infrastructure as a Service (IaaS)
 - Explain edge computing